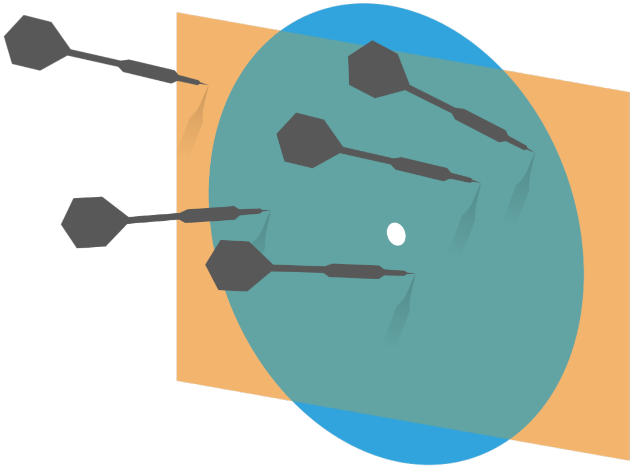
Probability and Statistics for Computer Science



Credit: wikipedia

"I have now used each of the terms mean, variance, covariance and standard deviation in two slightly different ways." ---Prof. Forsythe

Hongye Liu, Teaching Assistant Prof, CS361, UIUC, 09.19. 2024

Last time

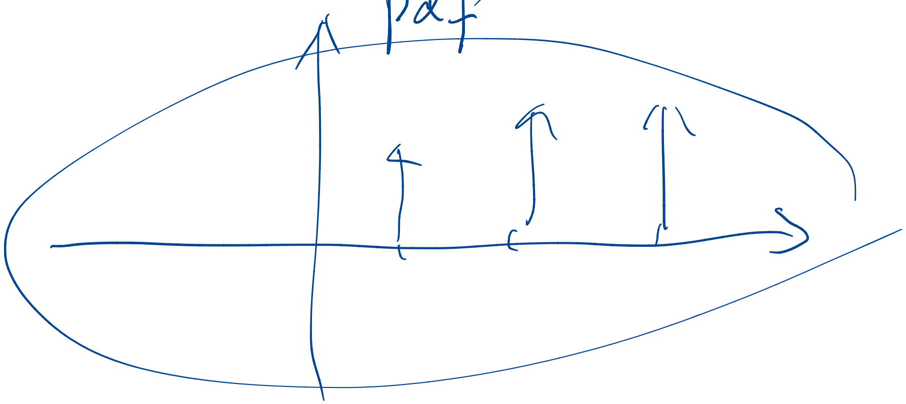
- Random Variable
- Probability distribution (pdf)

xx function

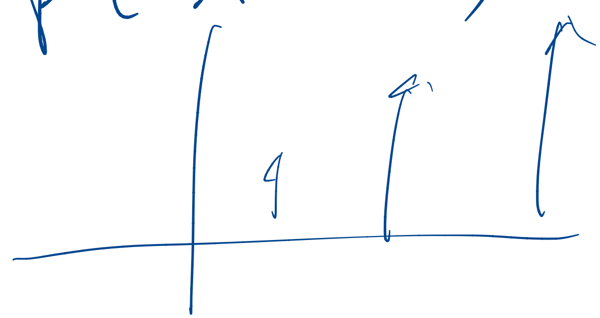
- Cumulative distribution (cdf)

$P(X = x)$

pdf



$P(X \leq x)$



Objectives

- Random Variable
  - Conditional probability and joint probability
  - *Independence of random variables*
  - *Expected value*
  - *Variance & covariance*

Two RVs have the same identical pdf

1. They have the same possible values

2. They have the same corresponding probabilities.

$\Rightarrow$ They have identical pdf plots!!

flip of a fair coin

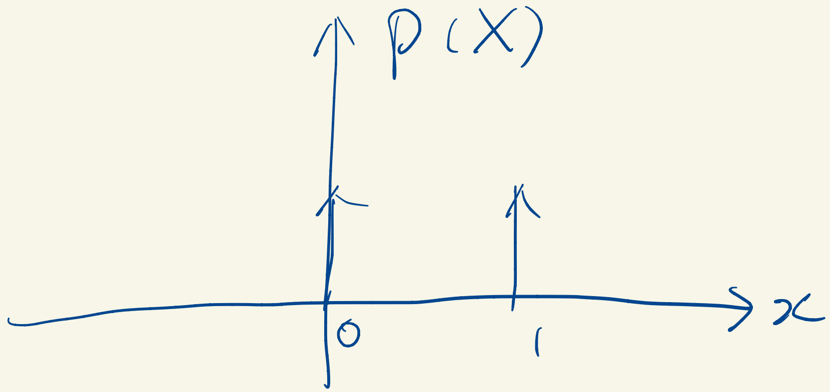
$$X(\omega) = \begin{cases} 1 & \text{Head} \\ 0 & \text{Tail} \end{cases}$$

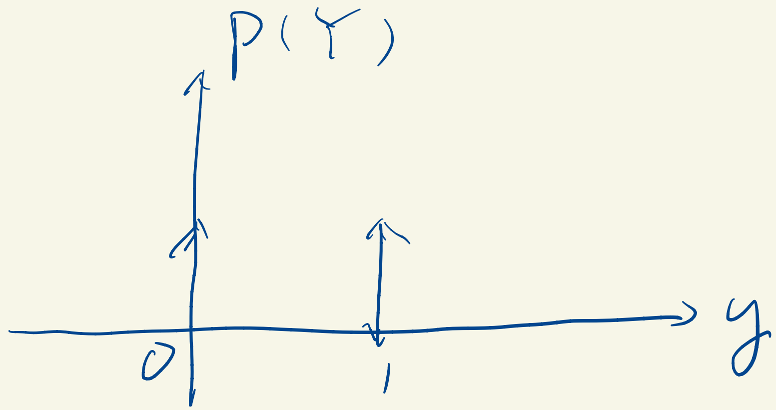
$\omega \in \Omega_1$

roll of a fair 4-die

$$Y(\omega) = \begin{cases} 1 & \text{even} \\ 0 & \text{odd} \end{cases}$$

$\omega \in \Omega_2$





they have the same pdf

probility distribution func.

Two RVs are equal

$$RV_2 = RV_1$$

$$\hookrightarrow RV_2 = f(RV_1) = 1 \times RV_1$$

① They share the same experiment.

② They have identical pdf.

If RV_1 takes a value v_1 after the experiment

RV_2 also takes the same value.

RV as function of two RVs

$P(X=1) = 3/4$

$P(Y=1) = 3/4$

$$S = X + Y$$

X, Y indept

$$X = \begin{cases} 1 & H \\ 0 & T \end{cases} \qquad Y = \begin{cases} 1 & H \\ 0 & T \end{cases}$$

$P(X=0, Y=1)$

$P(S=1) = ?$

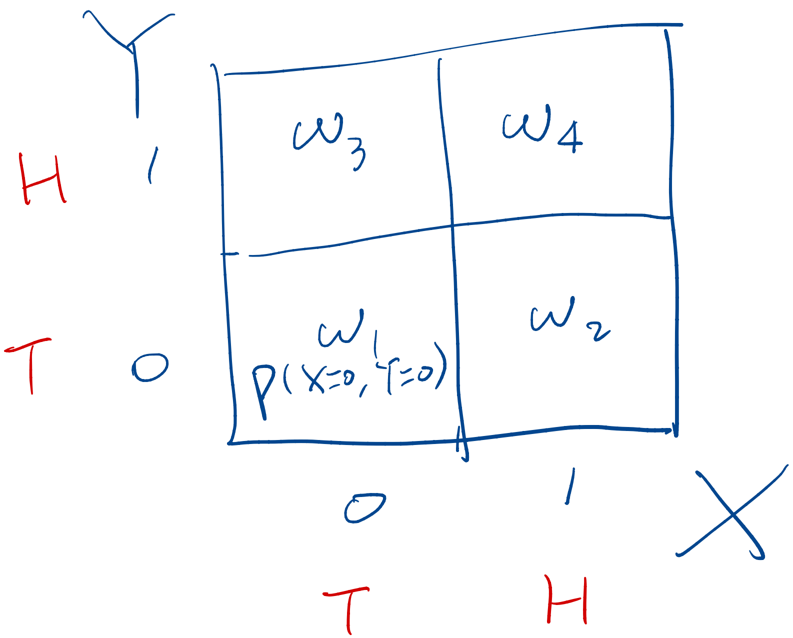
Y \ X	0	1
1	1	2
0	0	1

$S \quad \Omega \to \{0 \quad 1 \quad 2\}$

$P(X=1, Y=0) = P(X=1)\,P(Y=0)$

$3/4 \times 1/4$

$$f(X,Y) = X+Y$$



$$V = f(X,Y)$$

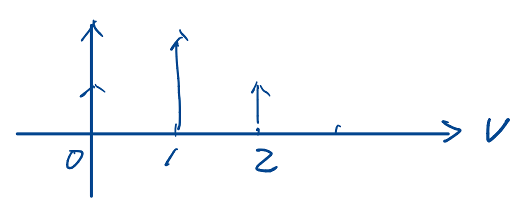
$\omega_1 \to 0+0$

$\omega_2 \to 1+0$

$\omega_3 \to 0+1$

$\omega_4 \to 1+1$

$p(V)$



Function of random variables: die example

Roll 4-sided fair die twice.

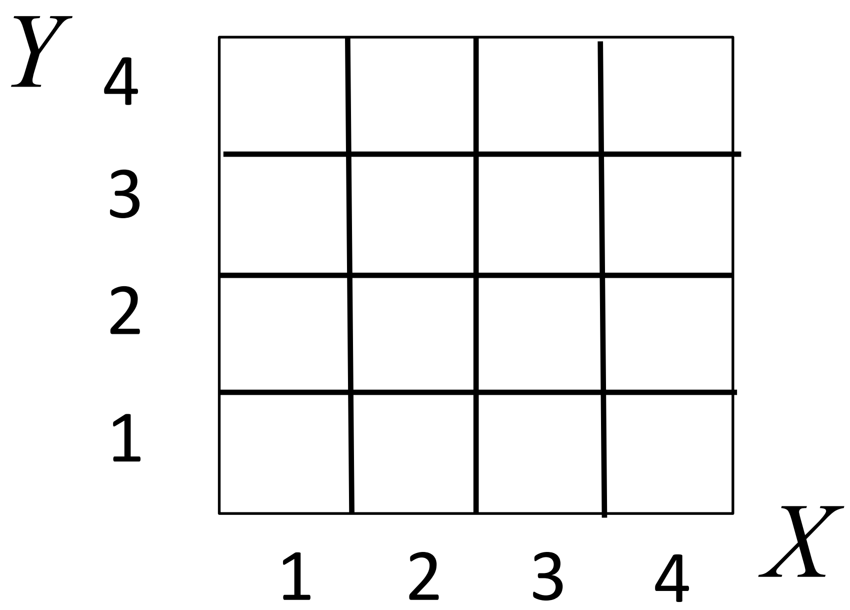
Define these random variables:

X, the values of 1st roll

Y, the values of 2nd roll

Sum $S = X + Y$

Difference $D = X - Y$



Random variable: die example

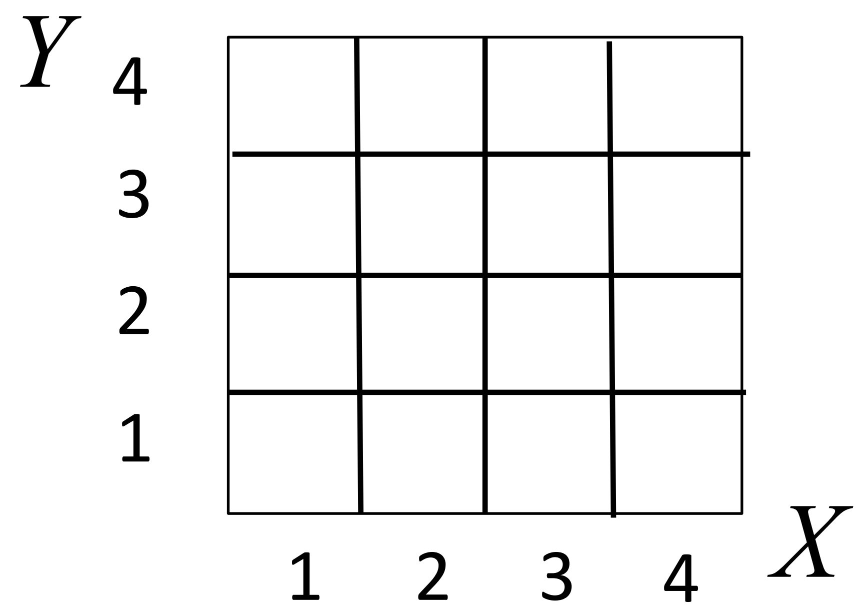
Roll 4-sided fair die twice. independently

$P(X = 1)$ = 1/4

$P(Y \leq 2)$ = 1/2

$P(S = 7)$ = 2/16

$P(D \leq -1)$



Random variable: die example

$S = X + Y$

Y				
4	5	6	7	8
3	4	5	6	7
2	3	4	5	6
1	2	3	4	5
	1	2	3	4 X

$D = X - Y$

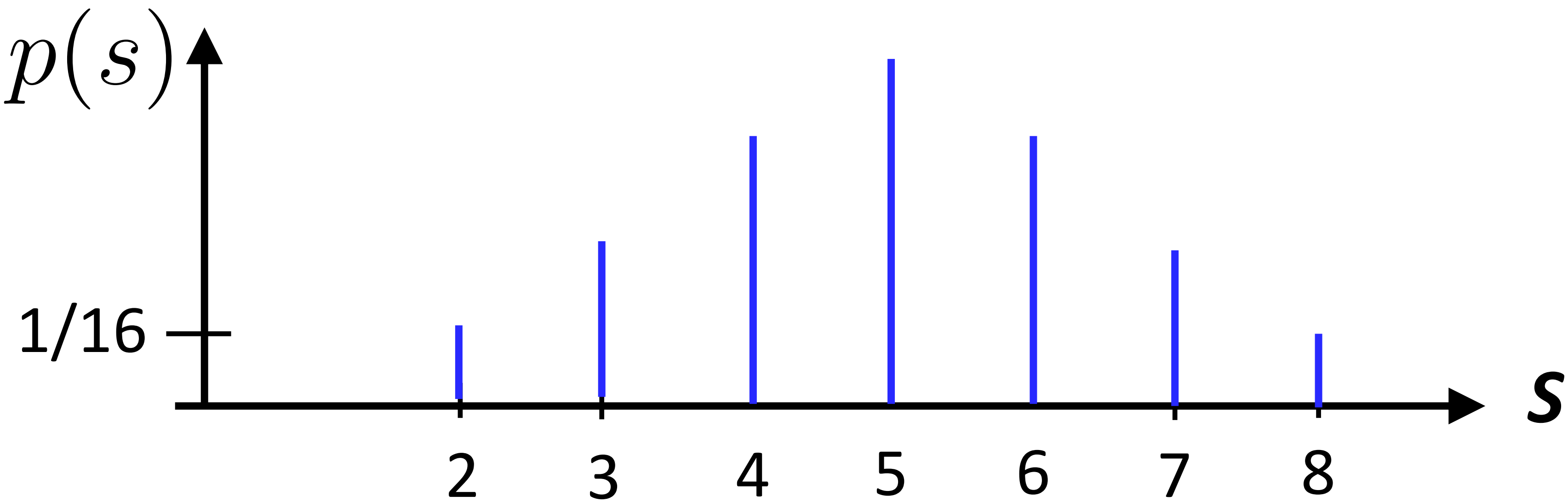
Y				
4	-3	-2	-1	0
3	-2	-1	0	1
2	-1	0	1	2
1	0	1	2	3
	1	2	3	4 X

$P(S = 7)$

$P(D \leq -1)$ $\frac{6}{16}$

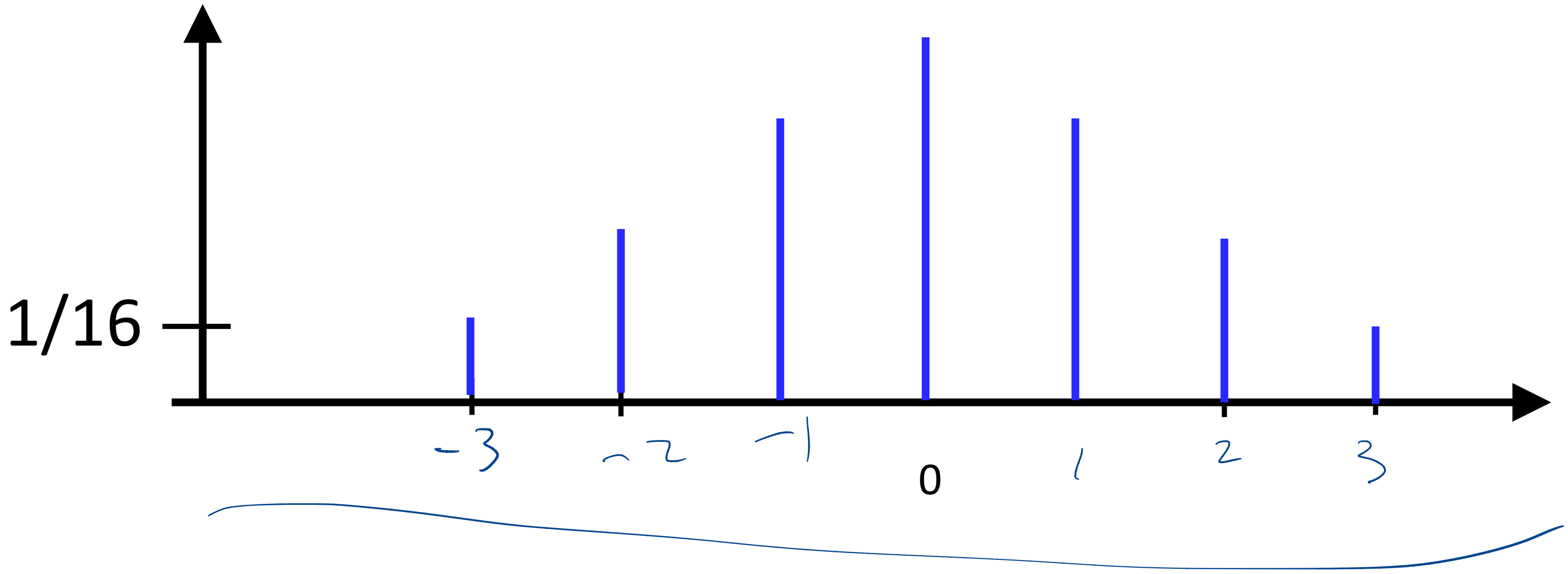
Probability distribution of the sum of two random variables

- Give the random variable ***S*** *in the 4-sided die, whose range is* {2,3,4,5,6,7,8}, probability distribution of ***S*.**



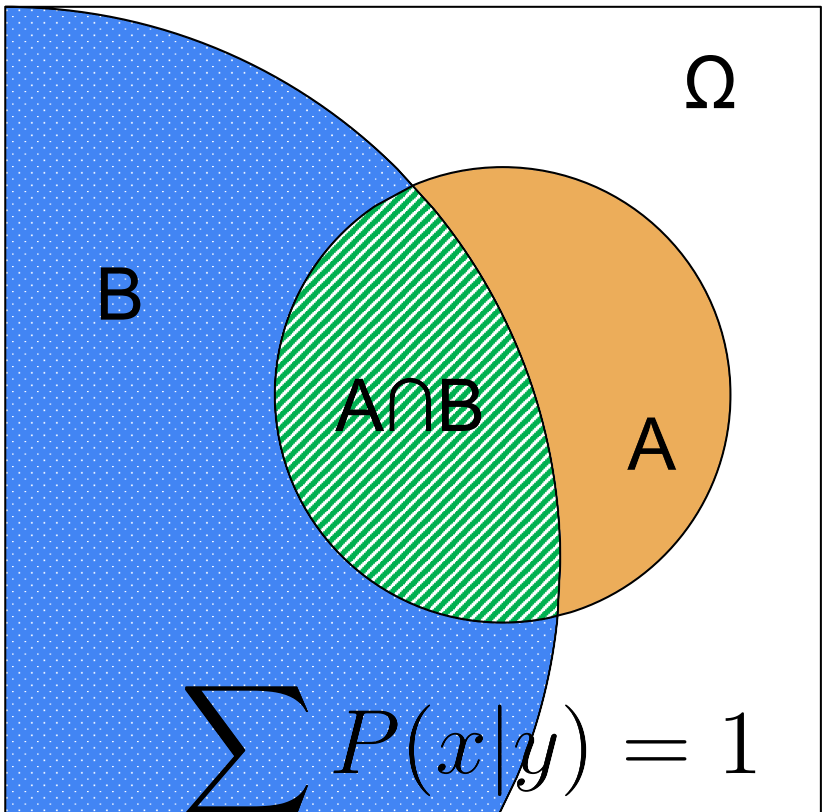
Probability distribution of the difference of two random variables

- Give the random variable ***D = X-Y***, *what is the probability distribution of* ***D****?*



Conditional Probability

- The probability of ***A*** given ***B***



$$\sum_x P(x|y) = 1$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) \neq 0$$

The “Size” analogy

Conditional probability distribution of random variables

- The conditional probability distribution of $X\ given\ Y$ is

$$P(x|y) = \frac{P(x,y)}{P(y)} \qquad P(y) \neq 0$$

$$p(x|y) = \frac{p(X=x \cap Y=y)}{p(y)}$$

$p(y) \rightarrow p(Y=y)$

$\rightarrow p(X=x|Y=y)$

Conditional probability distribution of random variables

- The conditional probability distribution of $X\ given\ Y$ is

$$P(x|y) = \frac{P(x,y)}{P(y)} \quad P(y) \neq 0$$

- The joint probability distribution of two random variables ***X*** and ***Y*** is

$$P(\{X = x\} \cap \{Y = y\})$$

$$\sum_x P(x|y) = 1$$

$\sum_x P(x|y) = 1$

Get the marginal from joint distri.

- We can recover the individual probability distributions from the joint probability distribution

$$P(x) = \sum_y P(x, y)$$

$$P(y) = \sum_x P(x, y)$$

$$p(x, y) = p(y|x)\, p(x)$$

$$\sum_y p(y|x)\, p(x)$$

$$p(x) \cancelto{1}{\sum_y p(y|x)}$$

$$= p(x)$$

Joint probabilities sum to 1

- The sum of the joint probability distribution

$$\sum_y \sum_x P(x, y) = 1$$

Joint Probability Example

- Tossing a fair coin twice, we define random variable $X \, and \, Y$ for each toss.

$$X(\omega) = \begin{cases} 1 & outcome \ of \ \omega \ is \ head \\ 0 & outcome \ of \ \omega \ is \ tail \end{cases}$$

$$Y(\omega) = \begin{cases} 1 & outcome \ of \ \omega \ is \ head \\ 0 & outcome \ of \ \omega \ is \ tail \end{cases}$$

Joint probability distribution example

$P(x, y)$

		X = 0	X = 1
Y	0	$P(X=0, Y=0) = P(X=0) \times P(Y=0) = \frac{1}{2} \times \frac{1}{2}$	$\frac{1}{4}$
	1	$\frac{1}{4}$	$\frac{1}{4}$

$P(y)$

0	$\frac{1}{2}$
1	$\frac{1}{2}$

$P(x)$

0	1
$\frac{1}{2}$	$\frac{1}{2}$

$P(y=0) = \frac{1}{2}$

$P(y=1) = \frac{1}{2}$

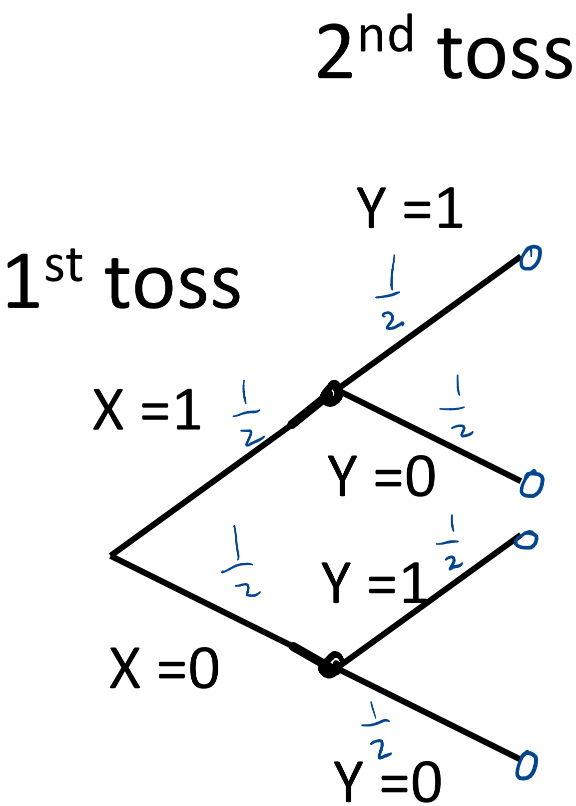
Joint Probability Example

Now we define Sum ***S*** = $X + Y$, Difference ***D*** = $X - Y$. ***S*** takes on values {0,1,2} and ***D*** takes on values {-1, 0, 1}

$$X(\omega) = \begin{cases} 1 & outcome\ of\ \omega\ is\ head \\ 0 & outcome\ of\ \omega\ is\ tail \end{cases}$$

$$Y(\omega) = \begin{cases} 1 & outcome\ of\ \omega\ is\ head \\ 0 & outcome\ of\ \omega\ is\ tail \end{cases}$$

Joint Probability Example



$S = X + Y$ $D = X - Y$

S	D	P(s, d)
2	0	$P(S=2, D=0) = \frac{1}{4}$
1	1	$P(S=1, D=1) = \frac{1}{4}$
1	-1	$P(S=1, D=-1) = \frac{1}{4}$
0	0	$P(S=0, D=0) = \frac{1}{4}$

Suppose coin is fair, and the tosses are independent

Joint probability distribution example

$P(s,d)$

S \ D	-1	0	1
0	0	$\frac{1}{4}$	0
1	$\frac{1}{4}$	0	$\frac{1}{4}$
2	0	$\frac{1}{4}$	0

$P(s)$

0	$\frac{1}{4}$
1	$\frac{1}{2}$
2	$\frac{1}{4}$

$P(d)$

-1	0	1
$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Independence of random variables

- Random variable $X\ and\ Y$ are independent if

$$P(x,y) = P(x)P(y)\ \boxed{for\ all\ x\ and\ y}$$

- In the previous coin toss example
  - Are X and Y independent? indpt
  - Are ***S*** and ***D*** independent?

Joint probability distribution example

$P(s,d)$

S \ D	-1	0	1	$P(s)$
0	0	$\frac{1}{4}$	0	$\frac{1}{4}$
1	$\frac{1}{4}$	$\boxed{0}$	$\frac{1}{4}$	$\boxed{\frac{1}{2}}$
2	0	$\frac{1}{4}$	0	$\frac{1}{4}$
$P(d)$	$\frac{1}{4}$	$\boxed{\frac{1}{2}}$	$\frac{1}{4}$	

$P(D=0, S=1) = 0$

$P(D=0) = \frac{1}{2}$

$P(S=1) = \frac{1}{2}$

Joint probability distribution example

$P(s, d)$

S \ *D*	-1	0	1
0	0	$\frac{1}{4}$	0
1	$\frac{1}{4}$	$\boxed{0}$	$\frac{1}{4}$
2	0	$\frac{1}{4}$	0

$P(s)$

0	$\frac{1}{4}$
1	$\boxed{\frac{1}{2}}$
2	$\frac{1}{4}$

$P(d)$

-1	0	1
$\frac{1}{4}$	$\boxed{\frac{1}{2}}$	$\frac{1}{4}$

$$P(S=1, D=0) \neq P(S=1)P(D=0)$$

Review at home

Conditional probability distribution example

$$P(s|d) = \frac{P(s,d)}{P(d)}$$

S \ D	-1	0	1
0	0	$\frac{1}{2}$	0
1	$\boxed{1}$	0	$\boxed{1}$
2	0	$\frac{1}{2}$	0

review at home

Bayes rule for random variable

- Bayes rule for events generalizes to random variables

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$\boxed{\begin{aligned} P(x|y) &= \frac{P(y|x)P(x)}{P(y)} \\ &= \frac{P(y|x)P(x)}{\sum_x P(y|x)P(x)} \end{aligned}}$$

Total Probability

review at home

Conditional probability distribution example

$$P(s|d) = \frac{P(s,d)}{P(d)}$$

S \ D	-1	0	1
0	0	$\frac{1}{2}$	0
1	1	0	1
2	0	$\frac{1}{2}$	0

$$P(D=-1|S=1) = \frac{P(S=1|D=-1)P(D=-1)}{P(S=1)} = \frac{1 \times \frac{1}{4}}{\frac{1}{2}}$$

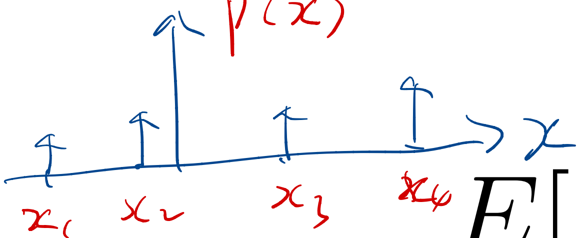
Three important facts of Random variables

- Random variables have **probability functions**
- Random variables can be **conditioned** on events or other random variables
- Random variables have **averages**

Expected value

- The **expected value** (or **expectation**) of a random variable X is

$0 \le p(x) \le 1$



$$E[X] = \sum_x x P(x)$$

$\sum_x p(x) = 1$

The expected value is a weighted sum of **all** the values X can take

Expected value

- The **expected value** of a random variable X is

<= 1

$$E[X] = \sum_x x \boxed{P(x)}$$

The expected value is a weighted sum of all the values X can take

Expected value: profit

- A company has a project that has ***p*** probability of earning 10 million and ***1-p*** probability of losing 10 million.
- Let X be the return of the project.

$$E[X] = ? \qquad \sum_x x\, p(x)$$

$$x : 10, \text{ or } -10$$

$$= 10 \cdot p + (-10)(1-p)$$

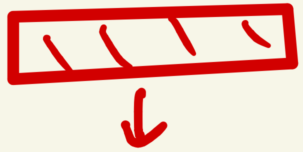
$$= 20p - 10 > 0$$

After class

Cookie

\$1 each

chocolate



\$2 each

A) random draw 1 out of 4 Expected value?

B) random draw 1 **twice** with replacement

when the two are the same, you get the prize.

Expected value?

Linearity of Expectation

- For random variables X and Y and constants k,c
  - Scaling property

$$E[kX] = kE[X]$$

  - Additivity

$$E[X+Y] = E[X] + E[Y]$$

  - And $E[kX+c] = kE[X] + c$

$f(aX+b)$
$= af(x)+b$
$f(ax+bY)$
$=af(x)+bf(Y)$

Linearity of Expectation

- Proof of the additive property

$$E[X+Y] = E[X] + E[Y]$$

$S = X + Y$

$$E[X+Y] = E(S) = \sum_{s} s\,P(s)$$

$$= \sum_{\{S=x+y\}} s \sum_{\{S=x+y\}} P(x,y)$$

$$= \sum_{x} \sum_{y} (x+y)\,P(x,y)$$

$P(s) = P(S=s)$

Note. $P(S=s) = 0$ if $x+y \neq s$

Proof conti.

$$E[X+Y] = \sum_x \sum_y (x+y)P(x,y)$$

$$= \sum_x \sum_y xP(x,y) + \sum_x \sum_y yP(x,y)$$

$$= \sum_x x \sum_y P(x,y) + \sum_y \sum_x y P(x,y)$$

$$= \sum_x x \cdot P(x) + \sum_y y \sum_x P(x,y)$$

$$= E[X] + \sum_y y P(y)$$

$$= E[X] + E[Y]$$

X, Y, Z, the same pdf

$$p(x) = \begin{cases} p & x=1 \\ 1-p & x=0 \\ 0 & ow \end{cases}$$

otherwise

$$E[X+Y-2Z]$$

$$= 0$$

$$E[X] = \sum x\,p(x)$$

$$= 1 \times p + 0 \times (1-p) = p$$

$$= E[X] + E[Y] - 2E[Z]$$

$$E[Y]$$

$$E[Z]$$

$$= 0$$

$$E[X] = E[Y] = E[Z]$$

$\because$ they have the same pdf.

Q. What's the value?

- What is $E[E[X]+1]$?

$\boxed{\text{A.}}$ $E[X]+1$ B. 1 C. 0

$$\Sigma\, y\, P(y)$$

$$(E[X]+1) \times 1$$

Expected value of a function of X

- If $\boldsymbol{f}$ is a function of a random variable X, then $Y = \boldsymbol{f}(X)$ is a random variable too

X^2

- The expected value of $Y = \boldsymbol{f}(X)$ is

$$E[Y] = E[f(X)]$$

$$= \sum_y y\, P(Y=y)$$

Expected value of a function of X

- If $\boldsymbol{f}$ is a function of a random variable X, then $Y = \boldsymbol{f}(X)$ is a random variable too
- The expected value of $Y = \boldsymbol{f}(X)$ is

$$E[Y] = E[f(X)] = \sum_x f(x)P(x)$$

i.e. $f(X) = X^2 \quad E[X^2] = \sum_x x^2 \cdot P(x)$

The exchange of variable theorem

$E[f(x)] = E[Y] = \sum y\, p(y)$

if $\{x\} \to \{y\}$ is bijection $\quad p(y) = p(x)$

$= \sum y\, p(x)$

$= \sum_x f(x)\, p(x)$

if several x in $\{x_s\} \to$ one y_s value

$$E[Y] = \sum_{y \in 1} y\, p(x) + y_s\, p(y_s)$$

$$= \sum_{y \in 1} y\, p(x) + y_s \sum_{x \in \{x_s\}} p(x)$$

$$= \sum y\, p(x) = \sum_x f(x)\, p(x)$$

Expected time of cat

- A cat moves with random constant speed **V**, either 5mile/hr or 20mile/hr with equal probability, what's the expected time for it to travel 50 miles?

$$T = \frac{50}{V}$$

$$E[V] = 5 \times \frac{1}{2} + 20 \times \frac{1}{2}$$

$$\underline{E[f(x)] \neq f(E[x])}$$

$$E[T] = \sum_v f(v)\, p(v)$$

$$= \sum_v \frac{50}{v} \cdot p(v)$$

$$= \frac{50}{5} \times \frac{1}{2} + \frac{50}{20} \times \frac{1}{2}$$

ans: 6.25

Review at home

Q: Is this statement true?

If there exists a constant such that $P(X \geq a) = 1$, then $E[X] \geq a$. It is:

$$E[X] = \sum_{x} x\,p(x)$$

A. True

B. False

$$= \sum_{x \geq a}^{\infty} x\,p(x) + \sum_{-\infty}^{\text{"}a\text{"}} x\,\cancelto{0}{p(x)}$$

$$= \sum_{x \geq a}^{\infty} x\,p(x) \geq \sum_{x \geq a}^{\infty} a\,p(x) = a$$

Variance and standard deviation

- The variance of a random variable X is

$$\boxed{var[X] = E[(X - E[X])^2]}$$

- The standard deviation of a random variable X is

$$std[X] = \sqrt{var[X]}$$

$E[f(X)]$

$= \sum_x f(x) \cdot p(x)$

$= \sum_x (x - E[X])^2 \cdot p(x)$

Properties of variance

- For random variable X and constant k

$$var[X] \geq 0$$

$$var[kX] = k^2 var[X]$$

$$var[X+c] = var[X]$$

Assignments

- Finish week4 module
- Next time: Markov and Chebyshev inequality & Weak law of large numbers, Continuous random variable

Additional References

- Charles M. Grinstead and J. Laurie Snell "Introduction to Probability"
- Morris H. Degroot and Mark J. Schervish "Probability and Statistics"

See you next time

See You!